

## *Cables and Suspension Bridges*

---

### 15.1. INTRODUCTION

---

Cables are commonly used to support suspension bridges, building roofs, as permanent guys for transmission towers and chimneys, and as temporary guys during erection. These are more suitable over the beam and truss, for spans greater than 50 m. Cables are usually flexible and carry their loads in tension only. Since cables, are always in tension, these are most economical and also will not become unstable and collapse, as may happen with beams and trusses.

Cables are suspended between supports and carry vertical loads. When a cable supports a load that is uniform per unit length of the cable itself, *e.g.* its self weight, it takes the form of a catenary. The analysis of catenary is complicated. The analysis is greatly simplified by assuming the uniform load distributed along the horizontal span, resulting in a parabolic shape. The assumption made is justified till the sag of the cable is not large in proportion to its length. The central dip of the cable varies from  $1/10$  to  $1/12$  of the span ; greater the dip, less is the horizontal thrust. In case, if a number of concentrated loads act on the cable the resulting profile will be a funicular polygon, the shape depending upon the position of loads (Fig. 15.1).

Suspension cable bridges are very light and are used for pedestrian traffic only. Cable with stiffening girders are used for roadways. Due to oscillations and sag under rolling loads, these types of bridges are not suitable for railways.

### 15.2. ANALYSIS OF CABLES UNDER CONCENTRATED LOADS

---

(i) *Cable suspended from supports at same level.* Consider a cable supported at 'A' and 'B',  $l$  distance apart, and subjected to loads  $W_1, W_2, W_3$  and  $W_4$  at distances  $x_1, x_2, x_3$  and  $x_4$  respectively from 'A'. Under the loads the cable will take up the shape of funicular polygon  $ACDEFB$ . The reactions at 'A' will be  $R_A$  vertical and  $H_A$  horizontal. Similarly reactions at 'B' will be  $R_B$  vertical and  $H_B$  horizontal. Three equations of statics give,

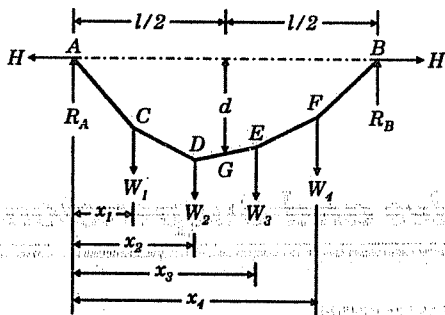


Fig. 15.1

$$H_A = H_B = H \quad \dots(1)$$

$$R_A + R_B = W_1 + W_2 + W_3 + W_4 \quad \dots(2)$$

Taking moments at 'A',

$$R_B \times l = W_1 x_1 + W_2 x_2 + W_3 x_3 + W_4 x_4 \quad \dots(3)$$

The fourth equation is obtained from consideration that the B.M. at any point in the cable is zero.

If the dip at the centre of span is 'd', equating B.M. at centre of a span equal to zero,

$$R_A \frac{l}{2} - H \times d - H_1 \left( \frac{l}{2} - x_1 \right) - W_2 \left( \frac{l}{2} - x_2 \right) = 0 \quad \dots(4)$$

These four equations give values  $H_A$ ,  $H_B$ ,  $R_A$  and  $R_B$ .

In case the dip at any other section at known distance from the support is given, fourth equation is obtained by putting B.M. at that point to zero.

The dip of the cable at point of application of load  $W_1$  is obtained by equating B.M. at that point to zero.

$$\therefore R_A x_1 - H y_1 = 0$$

$$y_1 = \frac{R_A x_1}{H}$$

Similarly dip at point of application of  $W_2$  can be obtained from

$$R_A x_2 - H y_2 - W_1 (x_2 - x_1) = 0$$

$$y_2 = \frac{R_A x_2 - W_1 (x_2 - x_1)}{H}$$

Knowing dip at various points the length of the cable can be calculated.

Maximum tension in the cable will be at the points of suspension.

$$\text{Tension at } A = \sqrt{R_A^2 + H^2}$$

$$\text{Tension at } B = \sqrt{R_B^2 + H^2}.$$

(ii) *Cable suspended from supports at different levels.* Consider cable supported at 'A' and 'B'. Let 'h' be the difference of levels between 'A' and 'B'. Let 'd' be the dip of the cable at point 'C', distance 'a' from the support 'A'. Let span  $AB = l$ .

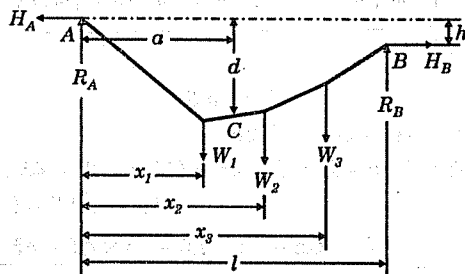


Fig. 15.2

$$\Sigma H = 0, \text{ gives } H_A = H_B = H \quad \dots(1)$$

$$\Sigma V = 0, \text{ gives } R_A + R_B = W_1 + W_2 + W_3 \quad \dots(2)$$

Taking moments about 'A',

$$R_B l + H_B h - W_1 x_1 - W_2 x_2 - W_3 x_3 = 0 \quad \dots(3)$$

Equating B.M. at 'C' to zero,

$$\therefore R_A \times a - H_A d - W_1(a - x_1) = 0 \quad \dots(4)$$

These four equations will give values of  $R_A$ ,  $R_B$ ,  $H_A$  and  $H_B$ .

Tension at 'A' in cable will be  $\sqrt{R_A^2 + H^2}$

and at 'B' will be  $\sqrt{R_B^2 + H^2}$ .

**Example 15.1.** A light suspension bridge is constructed to carry a pathway 3 m broad over a channel 21 m wide. The pathway is supported by six equidistant suspension rods. The cable has central dip of 2.0 m. The total load on the platform is 10 kN/m<sup>2</sup>. Find the maximum tension in the cable.

**Sol.** The pathway is carried by two cables, load transmitted to each cable by each suspender

$$= \frac{3 \times 3}{2} \times 10 = 45 \text{ kN}$$

$$R_A = R_B = 45 \times 3 = 135 \text{ kN}$$



**Sol.** Let  $R_A$  and  $R_B$  be vertical reactions at 'A' and 'B' respectively. Horizontal reactions

$$H_A = H_B = H$$

$$R_A + R_B = 2 + 4 + 3 + 5 = 14 \text{ kN}$$

Taking moments about 'B',

$$R_A \times 20 - H \times 1 - 2 \times 16 - 4 \times 12 - 5 \times 8 - 3 \times 4 = 0$$

$$20R_A - H - 32 - 48 - 40 - 12 = 0$$

$$20R_A - H = 132 \quad \dots(1)$$

B.M. at 'C' is zero

$$\therefore R_A \times 4 - H \times 1 = 0$$

$$\therefore H = 4R_A \quad \dots(2)$$

Substituting in (1)

$$20R_A - 4R_A = 132$$

$$\therefore R_A = \frac{132}{16} = 8.25 \text{ kN}$$

$$R_B = 14 - R_A = 14 - 8.25 = 5.75 \text{ kN}$$

$$H = 4R_A = 4 \times 8.25 = 33 \text{ kN.}$$

Maximum tension will be in portion AC

$$T = \sqrt{(8.25)^2 + (33)^2} = 34.02 \text{ kN}$$

B.M. at point 'D' equal to zero, gives

$$R_A \times 8 - Hy_D - 2 \times 4 = 0$$

$$8.25 \times 8 - 33y_D - 8 = 0$$

$$\therefore y_D = \frac{58}{33}$$

B.M. at point 'E' equal to zero, gives

$$R_A \times 12 - Hy_E - 2 \times 8 - 4 = 0$$

$$\therefore 8.25 \times 12 - 33y_E - 16 - 16 = 0$$

$$\therefore y_E = \frac{67}{33} \text{ m.}$$

B.M. at point 'F' equal to zero, gives

$$R_A \times 16 - Hy_F - 2 \times 12 - 4 \times 8 - 5 \times 4 = 0$$

$$\therefore 8.25 \times 16 - 33y_F - 24 - 32 - 20 = 0$$

$$\therefore y_F = \frac{56}{33} \text{ m.}$$

Lengths of different portions of cable will be

$$AC = \sqrt{4^2 + 1^2} = 4.123 \text{ m.}$$

$$CD = \sqrt{4^2 + \left(\frac{58}{33} - 1\right)^2} = 4.071 \text{ m.}$$

$$DE = \sqrt{4^2 + \left(\frac{67}{33} - \frac{58}{33}\right)^2} = 4.009 \text{ m.}$$

$$EF = \sqrt{4^2 + \left(\frac{67}{33} - \frac{58}{33}\right)^2} = 4.01 \text{ m.}$$

$$FB = \sqrt{4^2 + \left(\frac{56}{33} - 1\right)^2} = 4.06 \text{ m.}$$

Total length of the cable

$$= 4.123 + 4.071 + 4.009 + 4.01 + 4.06 = 20.273 \text{ m.}$$

### 15.3. ANALYSIS OF CABLES UNDER UNIFORMLY DISTRIBUTED LOADS

(i) *Cable suspended from supports at some level.* Let 'd' be the dip of the cable to centre 'C' and cable be loaded with uniformly distributed load of intensity  $w/\text{unit length}$ .

$$R_A = R_B = \frac{wl}{2}$$

Consider equation of half of the cable. Let  $H_C$  be horizontal thrust at centre 'C'.  $H_A = H_C$

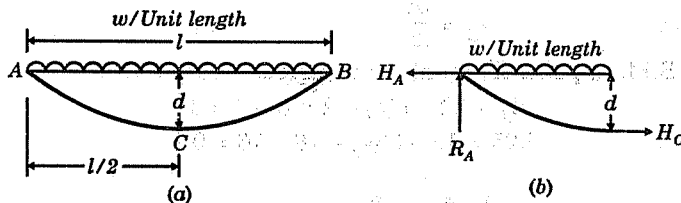


Fig. 15.5

B.M. at 'C' will be zero, therefore,

$$R_A \times \frac{l}{2} - H_A \times d - \frac{wl}{2} \times \frac{l}{4} = 0$$

$$\frac{wl}{2} \times \frac{l}{2} - H_A \times d - \frac{wl^2}{8} = 0$$

$$H_A = \frac{wl^2}{8d}$$

At centre of cable tension will be horizontal.

Let 'y' be sag at any point 'X', in the cable at distance 'x' from 'A'.

Taking moments about 'X',

$$R_A x - wx \times \frac{x}{2} - H_A y = 0$$

$$\therefore \frac{wl}{2} x - \frac{wx^2}{2} - \frac{wl^2}{8d} \times y = 0$$

$$y = \frac{8d}{l^2} \left[ \frac{dx}{2} - \frac{x^2}{2} \right] \quad \dots(1)$$

This represents a parabolic equation. Thus cable takes parabolic shape.

Maximum tension in the cable occurs at supports.

$$T_{max} = \sqrt{R_A^2 + H_A^2} = \sqrt{\left(\frac{wl}{2}\right)^2 + \left(\frac{wl^2}{8d}\right)^2}$$

$$= \frac{wl}{2} \sqrt{1 + \left(\frac{l}{4d}\right)^2} = \frac{wl}{8d} \sqrt{16d^2 + l^2}$$

### Length of the cable

$$\text{Length of the cable} = 2 \int_0^{l/2} ds$$

$$L = 2 \int_0^{l/2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\text{From (1)} \quad \frac{dy}{dx} = \frac{8d}{l^2} \left[ \frac{l}{2} - x \right]$$

$$\therefore L = 2 \int_0^{l/2} \sqrt{1 + \frac{64d^2}{l^4} \left(\frac{l}{2} - x\right)^2} dx$$

$$= 2 \int_0^{l/2} \left[ 1 + \frac{64d^2}{l^4} \left(\frac{l}{2} - x\right)^2 \right]^{\frac{1}{2}} dx$$

Expanding by binomial expansion and considering only first two terms

$$L = 2 \int_0^{l/2} \left[ 1 + \frac{1}{2} \times \frac{64d^2}{l^4} \left(\frac{l}{2} - x\right)^2 \right] dx$$

$$\begin{aligned}
 &= 2 \left[ x - \frac{32d^2}{l^4} \left( \frac{l}{2} - x \right)^3 \right]_0^{l/2} = 2 \left[ \frac{l}{2} - \frac{32d^2}{3l^4} \left\{ 0 - \left( \frac{l}{2} \right)^3 \right\} \right] \\
 &= 2 \left[ \frac{l}{2} + \frac{32d^2}{3l^4} \times \frac{l^3}{8} \right] = l + \frac{8}{3} \frac{d^2}{l}
 \end{aligned}$$

(ii) Cable suspended from supports at different levels and loaded uniformly with  $w$  / unit length.

Let 'h' be the difference of level between supports 'A' and 'B'.

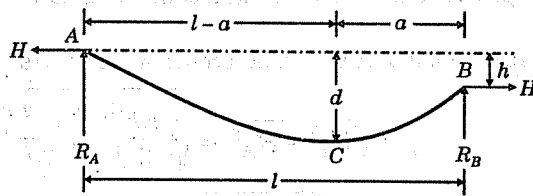


Fig. 15.6

Let dip of the cable be maximum at a point 'C', distance 'a' from 'B'.

$$H_A = H_B = H \quad \dots(1)$$

$$R_A + R_B = wl \quad \dots(2)$$

Taking moments about 'B',

$$R_A l - wl \times l/2 - H \times h = 0$$

$$\therefore R_A l - Hh = \frac{wl^2}{2} \quad \dots(3)$$

B.M. at 'C' is zero,

$$\therefore R_A(l-a) - Hd - \frac{w(l-a)^2}{2} = 0 \quad \dots(4)$$

These four equations can be used to evaluate four unknowns.

**Example 15.4.** A suspension cable, having supports at same level, has span of 30 m and maximum dip of 3 m. The cable is loaded with uniformly distributed load of 10 kN/m throughout its length and the concentrated load of 30 kN and 90 kN at middle third points. Find the maximum tension in the cable.

**Sol.** Taking moments about 'B'.

$$R_A \times 30 = \frac{30 \times 10 \times 30}{2} + 30 \times 10 + 90 \times 20$$



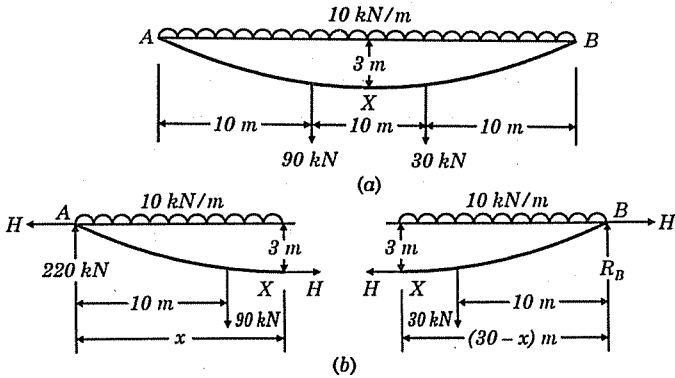


Fig. 15.7

$$R_A = 150 + 10 + 60 = 220 \text{ kN}$$

$$R_B = 300 + 90 + 30 - 220 = 200 \text{ kN}$$

Let maximum dip of 3 m be at distance 'x' from 'A'. At this section, tension in the cable will be horizontal and hence vertical component of tension in cable is zero.

Considering equilibrium of left portion of cable.

Equating sum of vertical forces to zero,

$$220 - w \times x - 90 = 0$$

$$220 - 10 \times x - 90 = 0$$

$$\therefore x = 13 \text{ m.}$$

Taking moments about A,

$$3H - 13 \times 10 \times \frac{13}{2} - 90 \times 10 = 0$$

$$\therefore 3H = 845 + 900$$

$$H = 281.7 + 300 = 581.7 \text{ kN}$$

Maximum tension in the cable will occur at A.

$$T_{\max} = \sqrt{(220)^2 + (581.7)^2} = 621.3 \text{ kN.}$$

**Example 15.4.** A suspension cable is supported at two points 20 m apart. The left support is 2 m above the right support. The cable is loaded with uniformly distributed load of 10 kN/m throughout the span. The maximum dip in the cable from left support level is 4 m. Find the maximum tension in the cable.

**Sol.** Let  $R_A$  and  $R_B$  be vertical reactions at supports 'A' and 'B'.

$$H_A = H_B = H \quad \dots(1)$$

$$R_A + R_B = 10 \times 20 = 200 \text{ kN} \quad \dots(2)$$

Taking moments about B.

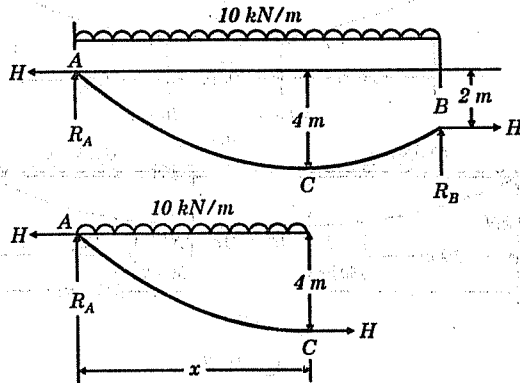


Fig. 15.8

$$H \times 2 - R_A \times 20 + 10 \times 20 \times 10 = 0$$

$$\therefore H - 10R_A + 1000 = 0 \quad \dots(3)$$

Let the lowest point of the cable 'C' be at distance 'x' from 'A'.

Consider free body diagram of AC. Taking moments about 'A',

$$H \times 4 - 10 \times x \times \frac{x}{2} = 0$$

$$\therefore H = \frac{10x^2}{8}$$

For portion AC,

$$\Sigma E = 0 \text{ gives, } R_A - 10 \times x = 0$$

$$\therefore R_A = 10x$$

Substituting values of H and  $R_A$  in terms of x in (3),

$$\frac{10x^2}{8} - 100x + 1000 = 0$$

$$x^2 - 80x + 800 = 0$$

$$(x - 40)^2 = 1600 - 800 = 800$$

$$x = 40 \pm 20\sqrt{2}$$

Neglecting +ve sign as x cannot be greater than 40 m.

$$x = 40 - 28.28 = 11.72 \text{ m.}$$

$$R_A = 10x = 117.2 \text{ kN}$$

$$H = \frac{10x^2}{8} = \frac{(117.2)^2}{8} \times 10 = 172 \text{ kN}$$

Maximum tension occurs at 'A',

$$T_{max} = \sqrt{R_A^2 + H^2} = \sqrt{(117.2)^2 + (172)^2} = 208.1 \text{ kN.}$$

#### 15.4. SHAPE OF CABLE UNDER SELF-WEIGHT

When a cable is suspended from two supports, it takes shape of catenary under self-weight. Consider a cable suspended from two supports. Let  $w$  be self-weight of cable per unit length.

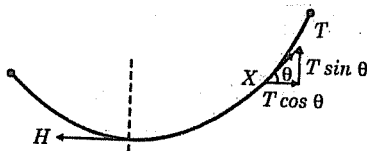


Fig. 15.9

Let ' $H$ ' be the tension in the cable at its lowest point and ' $T$ ' be tension in cable at some section ' $X$ '. Let ' $\theta$ ' be the slope of cable at this section considering equilibrium of right portion of cable.

$$H = T \cos \theta \quad \dots(1)$$

$$wS = T \sin \theta$$

where ' $S$ ' is the length of the cable between lowest point and ' $X$ '

From (1) 
$$T = \frac{H}{\cos \theta}$$

$$\therefore wS = \frac{H}{\cos \theta} \times \sin \theta$$

$$S = \int_0^x dx = \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$\therefore w \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \frac{H \times \sin \theta}{\cos \theta} = H \tan \theta$$

$$\tan \theta = \frac{dy}{dx}$$

$$\therefore w \int_0^x \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = H \frac{dy}{dx}$$

Differentiating both sides

$$w \sqrt{1 + \left(\frac{dy}{dx}\right)^2} = H \frac{d^2y}{dx^2}$$